



Figure 8: User login page of iAquarium

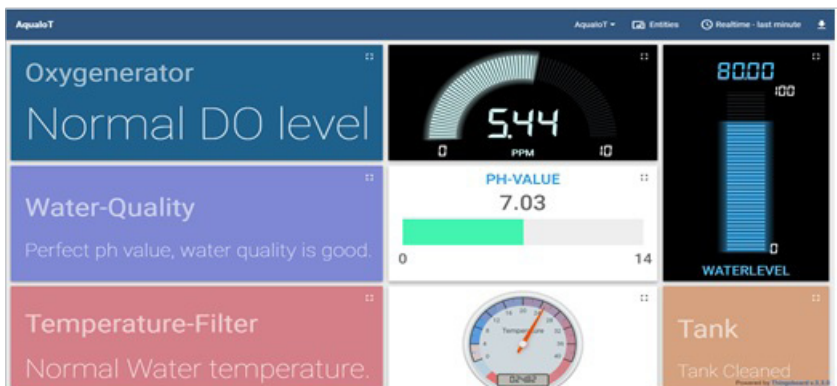


Figure 9: Dashboard visualisations of iAquarium

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which shows logs coming out of the container, and a terminal window as well to manually run a few commands if and when needed.

All the communication from CLI to the cloud happens over a secured API, which you can also use to develop custom applications for interacting with your fleets and devices.

Where to go from here

The example above is a very basic hello-world example, but there is a lot more that one can do. This includes interfacing sensors, displays, actuators; defining a multi-container app with networking privileges, and exposing ports.

balena has amazing docs which can be found at <https://www.balena.io/docs>. You can also check out balena

masterclasses for different balena tools at <https://www.balena.io/docs/learn/more/masterclasses/overview/>.

Two great projects made with balena, which you can try, are:

- balena Sense: <https://github.com/balenalabs/balena-sense>
- balena Dash: <https://github.com/balenalabs/balena-dash>

Some other great projects can be found at balenaHub, where community members have submitted their fleets, which you can join or fork. You can also learn about balena blocks, which are modular containers to ease functionalities like audio, Bluetooth connections, Chromium Web browser running from a container, and much more. You can check out all of this at hub.balena.io.

Web interface: This involves the creation of a Web interface, which consists of a login form to enter user details such as name, email ID, mobile number, and password. Once logged in, the user will be directed to view the status of the aquarium's tank. The UI design is simulated in Figure 7, and its outputs and dashboards are presented in Figures 8 and 9, respectively.

The idea here was to make a framework that can keep the fish in an aquarium healthy without carrying out manual monitoring of the water. This IoT based system enables consistent pH, water and temperature levels in an aquarium. This concept can be scaled for aquafarming. **END** 🐧

Dr Maheswari R.

The author is an associate professor at School of Computer Science and Engineering, VIT, Chennai.

You can also join the balena community on the following channels:

- balena forums: <https://forums.balena.io>
- balena blog: <https://balena.io/blog>
- balena labs GitHub: <https://github.com/balenalabs>

Hopefully, this tutorial will serve as a good starting point for anyone who wants to manage their IoT fleet efficiently. In case you have some questions, you can create a post on the balena forum, which is really active. **END** 🐧

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By: Ayan Pahwa

The author is an IoT developer advocate.